The Power of Vaccination

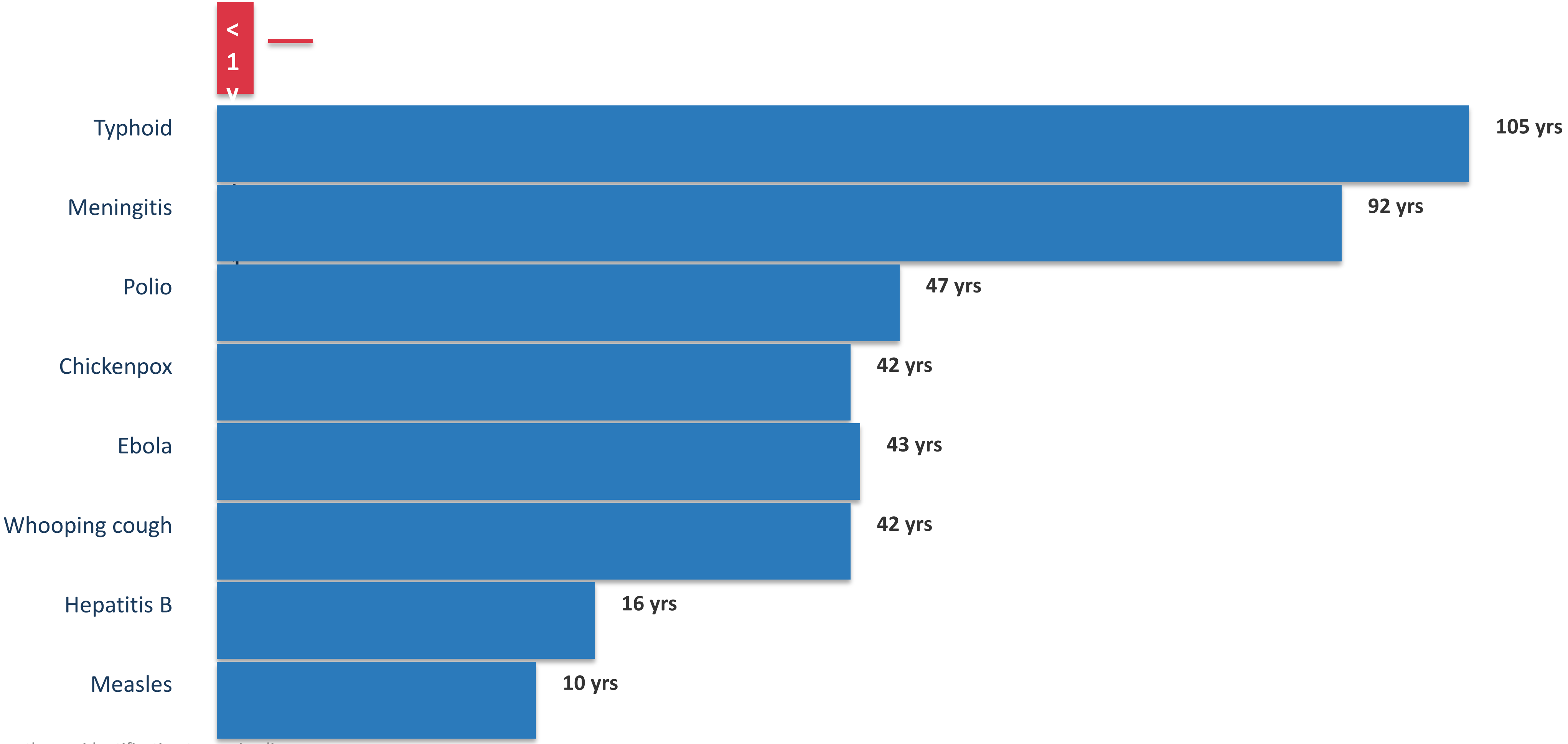
From Smallpox Eradication to COVID-19 in Record Time

1796–2020

Source: Our World in Data | Vaccination

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Years from pathogen identification to vaccine licensure



Source: Our World in Data — years from pathogen identification to vaccine licensure

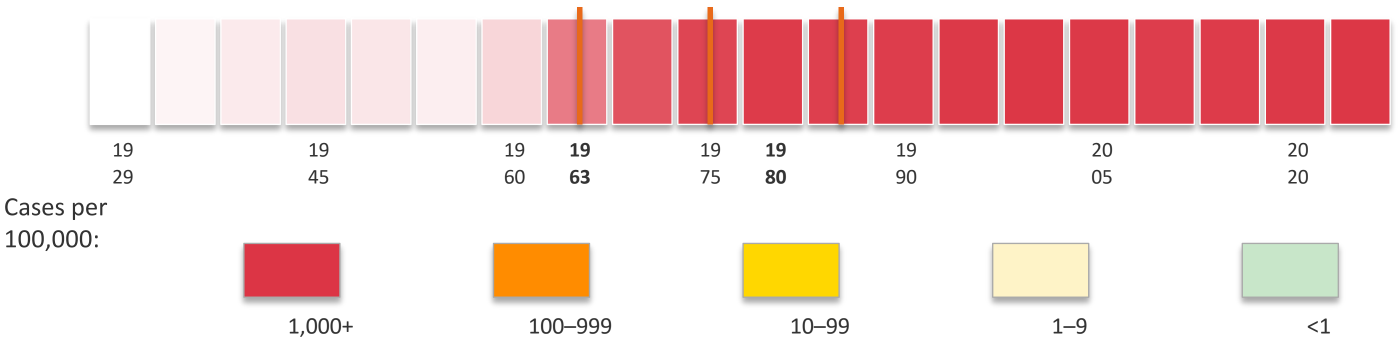
100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-Vaccine Cases (per million/yr)	Post-Vaccine Cases (per million/yr)	Case Reduction	Pre-Vaccine Deaths (per million/yr)	Post-Vaccine Deaths (per million/yr)	Death Reduction
Diphtheria	158	0	100%	13.7	0	100%
Measles	3,044	0.2	99.99%	2.5	0	100%
Smallpox	250	0	100%	2.9	0	100%
Acute polio	141	0	100%	10	0	100%
Pertussis	1,534	52	96.6%	30.8	0.1	99.7%
Rubella	242	0.04	99.98%	0.09	0	100%
Hepatitis A	465	51	89%	0.5	0.06	88.7%
Hepatitis B (acute)	273	44	83.9%	1	0.16	83.6%
Pneumococcal	233	139	40.5%	24	16.5	31.3%

- *Green shading = 100% case and death reduction*
- *Gray shading = reduction still incomplete*

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Measles cases per 100,000 population across US states, 1929–2022



BEFORE 1963	1980 MANDATE	POST-2000
Nearly every state reported 100–1,000+ cases per 100,000 annually	Cases dropped to near zero across all 50 states within a decade	Sporadic outbreaks only — most years below 1 case per 100,000

Smallpox: The Only Human Disease Eradicated Through Vaccination

1959
WHO passes resolution to eradicate smallpox

1967
Intensified global eradication campaign launched

1977
Last natural case in Somalia

1980
WHO certifies global eradication

Why Smallpox Was Eradicable:

Humans as only reservoir
The virus infected no animal populations — eliminating human cases meant eliminating the virus

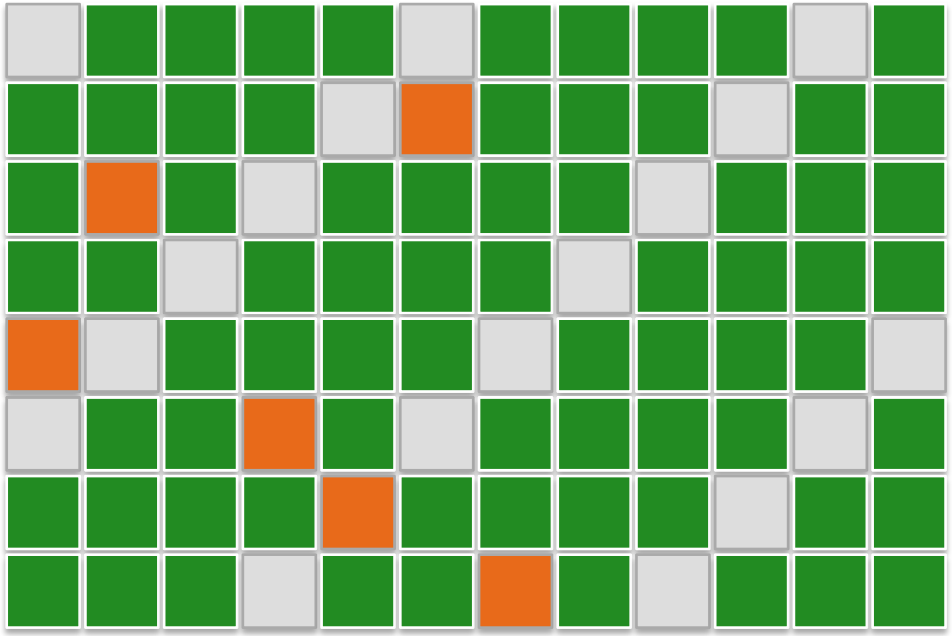
Distinguishable symptoms
The rash was visible and distinctive, enabling rapid case identification

Effective vaccine
Jenner's vaccine was cheap, stable, and provided decades of immunity

No asymptomatic carriers
Infectious individuals always showed symptoms, preventing silent transmission

Before eradication, smallpox killed an estimated 300 million people in the 20th century alone.

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



Vaccinated

Unvaccinated (healthy)

Vulnerable (cannot be vaccinated)

Pathogen blocked from spreading

When ≥80% of a population is vaccinated, pathogens cannot find new hosts — the chain of transmission breaks completely.

Shielding the unprotected

Newborns, immunocompromised patients, and cancer patients on chemotherapy depend on community immunity.

Herd threshold varies by disease

Highly contagious diseases like measles require ~95% coverage; less contagious diseases need ~80–85% coverage.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Pneumococcal Disease

Case reduction: 40.5%

Death reduction: 31.3%

Limited vaccine access for adults; serotype replacement

Hepatitis A

Case reduction: 89%

Death reduction: 88.7%

Incomplete childhood immunization coverage; outbreak vulnerability

Acute Hepatitis B

Case reduction: 83.9%

Death reduction: 83.6%

High-risk adult populations underserved; stigma around testing and treatment

Cross-Cutting Barriers to Higher Coverage:

Access Inequities

Socioeconomic and geographic barriers limit vaccine reach in underserved communities

Supply Shortages

Manufacturing constraints and distribution bottlenecks leave gaps in availability

Trust Deficits

Misinformation and low confidence reduce uptake even where vaccines are available

Key Takeaways

100%

100% Elimination Achieved

Diphtheria, smallpox, and both forms of polio have been eliminated in the US — cases and deaths reduced to zero through vaccination.

<1 yr

Development Timelines Collapsed

Vaccine development shrank from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.

Herd

Herd Immunity Shields Communities

When enough people are vaccinated, pathogens cannot spread, protecting newborns, immunocompromised, and chemotherapy patients.

Gaps

Coverage Gaps Persist

Pneumococcal disease (40.5%), hepatitis A (89%), and hepatitis B (83.9%) show incomplete control — access, supply, and trust remain barriers.

CALL TO ACTION: Improve access, coverage, and trust in vaccination to close remaining gaps and protect vulnerable populations worldwide.

Source: Our World in Data — ourworldindata.org/vaccination